LAB 05 — BUILDING A SCADA HUMAN MACHINE INTERFACE

Course: Intro to Critical Infrastructure for IT Learners

Module: Lab 05 — Building a SCADA Human Machine Interface

Institution: International Cybersecurity and Digital Forensics Academy (ICDFA)

Student Name: JOSEPH AGYAPONG

Registration ID: C1126ACIS17354

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Summary

This lab report demonstrates the development, configuration, and verification of an SCADA HMI (Human-Machine Interface) using FUXA software. The purpose of this project was to create a high-performance visual interface for a process control system using the OpenPLC Runtime simulation engine. The HMI was designed to operate in a real-time environment and follow situational-awareness principles, standard industry protocols such as Modbus TCP/IP, and the ISA-101 specification for high-performance HMIs.

Key Learning Outcomes Achieved:

- Applied high-performance HMI design principles to minimize operator cognitive load.
- Configured graphical process equipment (fluid storage tank, process piping, dynamic valve assembly, and bar level gauge) within FUXA.
- Configured Modbus TCP client-server communications between FUXA and OpenPLC Runtime.
- Bound discrete (Bool) and analog (Int16) tags to physical PLC addresses (%QX0.0 and %IW80).
- Validated dynamic visual state transitions (show/hide valve states and scaled 0–10 ft liquid tank levels).

2. Laboratory Environment & Architecture

The laboratory environment consists of two virtualized nodes operating on an isolated host-only network segment.

System Component	Hostname / Role	IP Address & Port	Credentials / Protocol
PLC Node	OpenPLC Runtime	192.168.1.10:502	plc / password (Modbus TCP)
SCADA Node	FUXA Web HMI	127.0.0.1:1881	scada / password (HTTP / WebSockets)

3. System Configuration & Tag Mapping

Modbus TCP Communication Parameters

- **Connection Name:** Open PLC
- **Target Address:** 192.168.1.10:502
- **Slave ID:** 1
- **Polling Interval:** 500 ms

Tag Addressing & Memory Mapping Table

Tag Name	Data Type	PLC Address	Modbus Register Type	Modbus Address / Offset	Scaling / Engineering Range
Tank Level	Int16	%IW80	Input Register (300080)	300080	0 – 10 ft

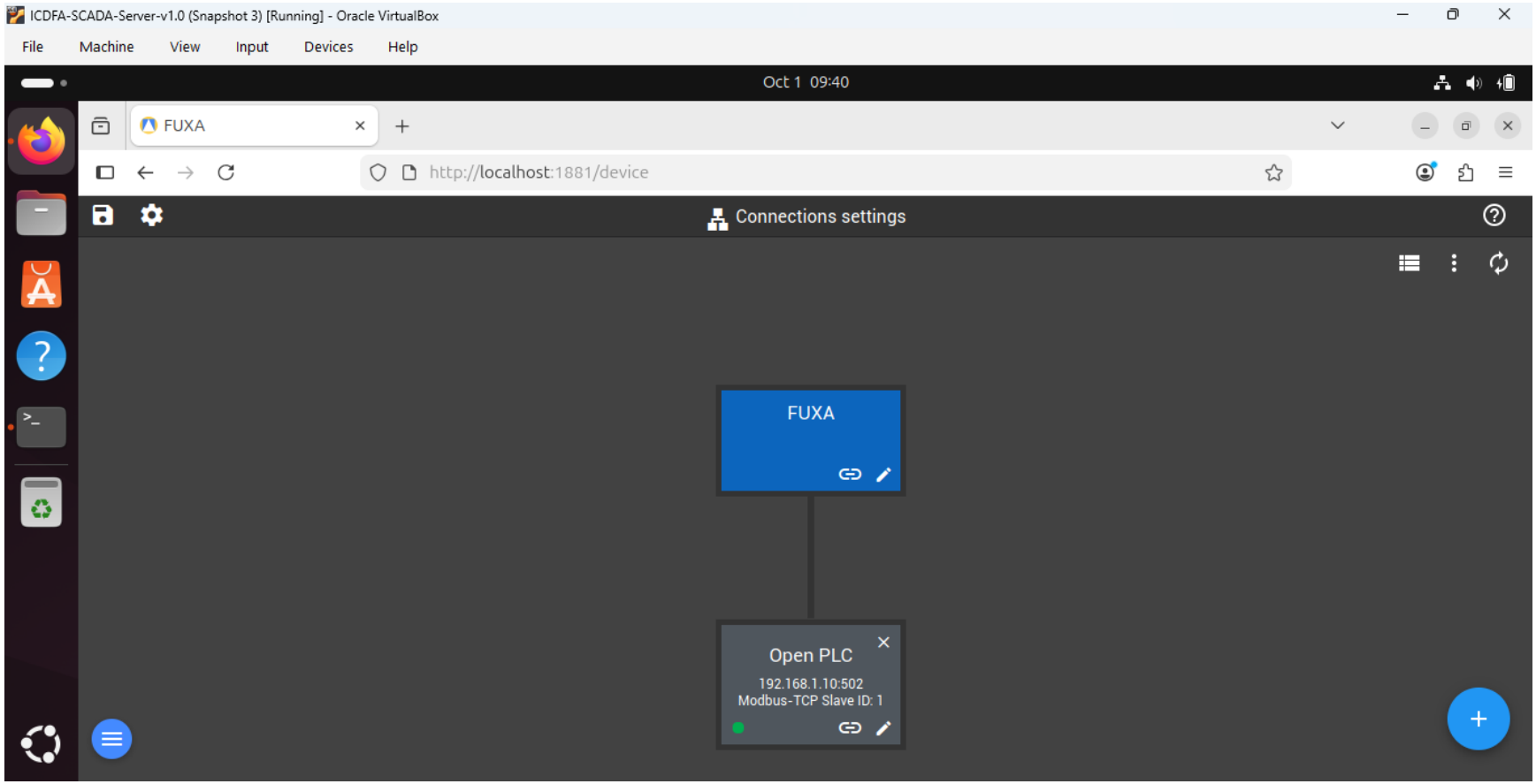
Tag Name	Data Type	PLC Address	Modbus Register Type	Modbus Address / Offset	Scaling / Engineering Range
Tank Valve Open	Bool	%QX0.0	Discrete Output / Coil	1 (1-based offset)	0 = Closed/Hide, 1 = Open/Show

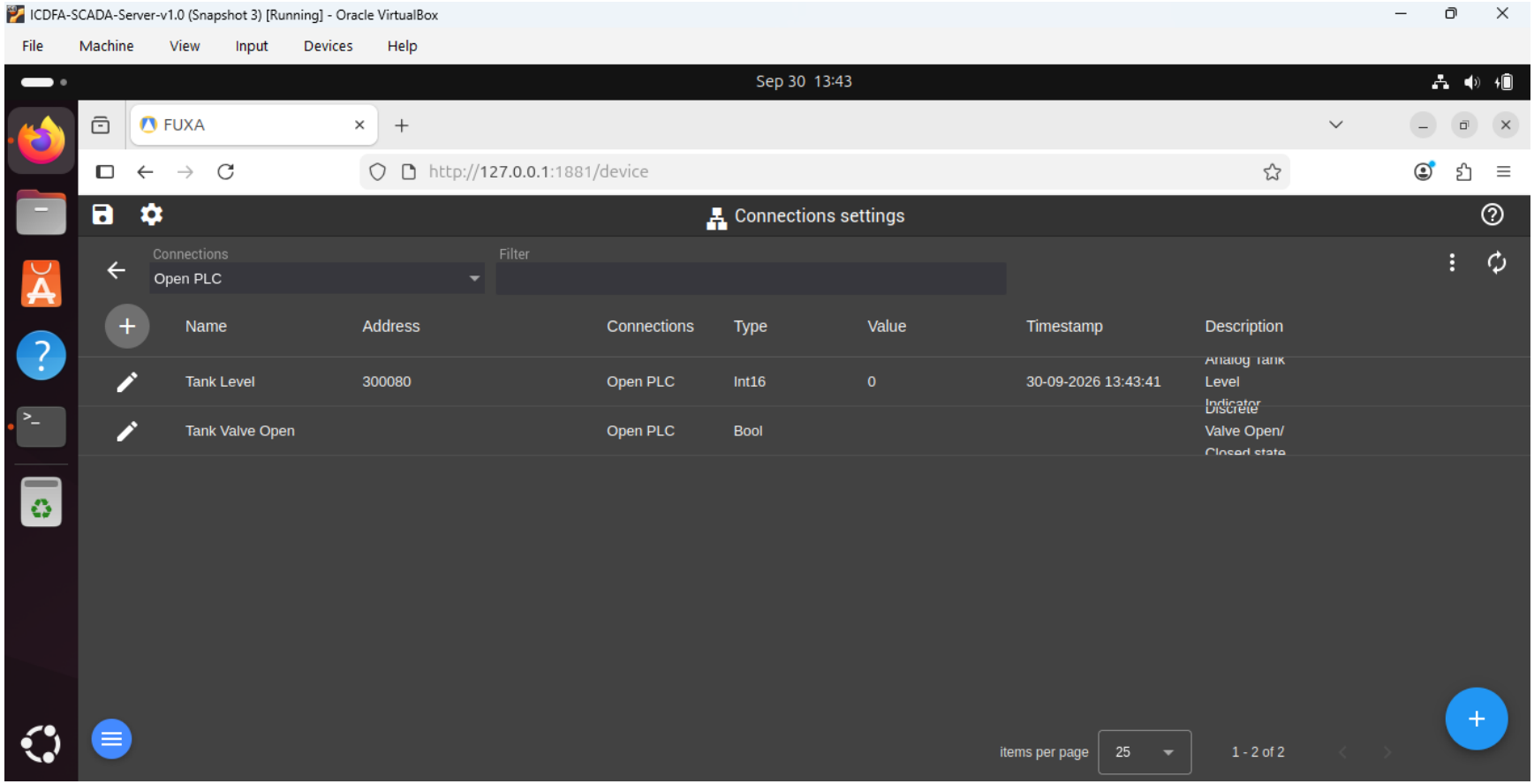
Dynamic Graphic Binding Rules

1. **Bar Gauge (Tank Fill):** Bound to Tank Level. Minimum value set to 0, Maximum value set to 10. Liquid height expands vertically as the register value increases.
2. **Fill Valve Assembly:** Grouped process graphic (Actuator, Valve Body, and Line). Bound to Tank Valve Open using dual-action visibility:
    - **Action 1:** Hide graphic when Tank Valve Open == 0 (FALSE).
    - **Action 2:** Show graphic when Tank Valve Open == 1 (TRUE).

4. Implementation Evidence & Screenshot Placeholders

Step 1: Modbus TCP Connection & Tag Mapping Verification

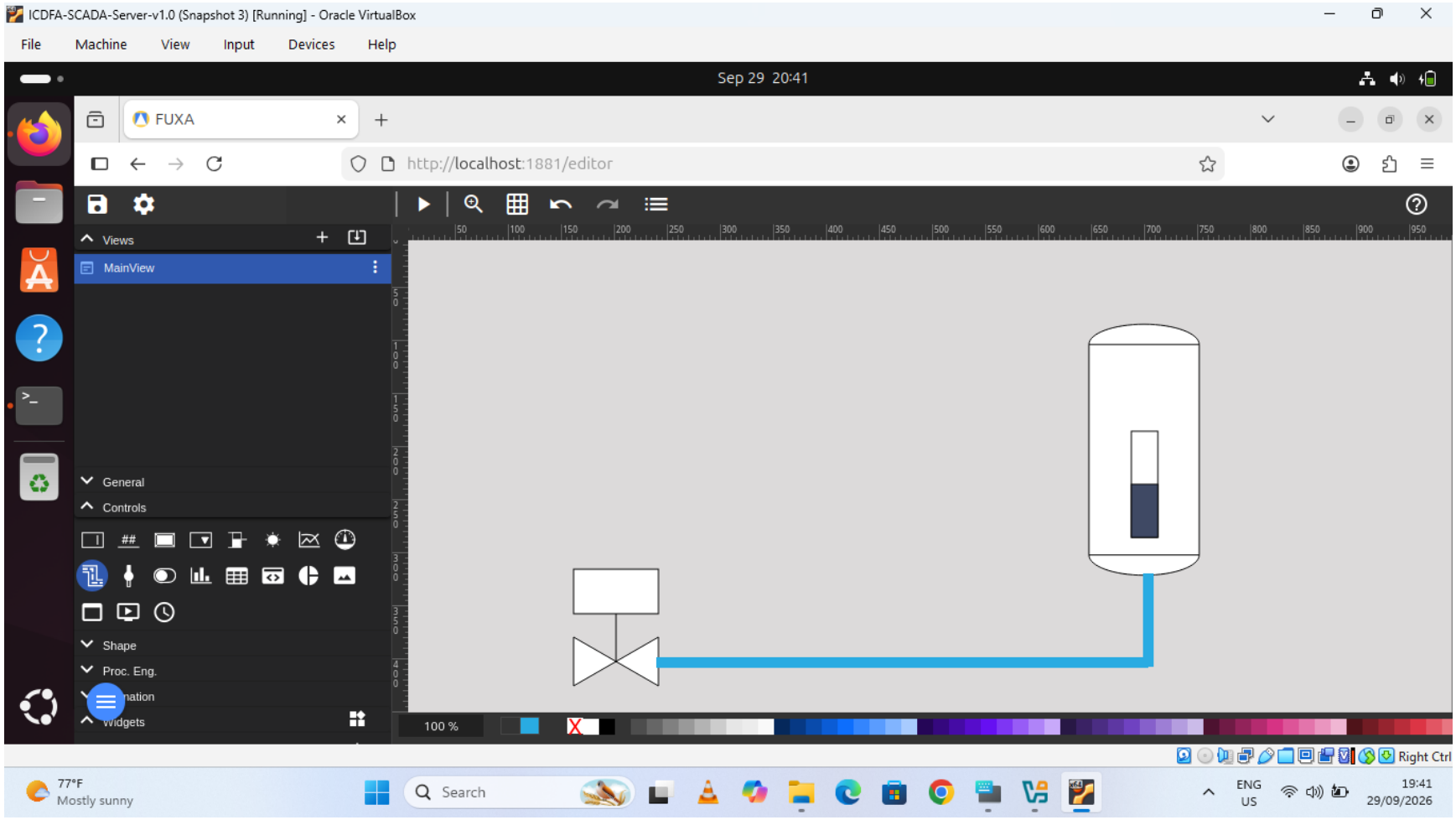

ICDFA-SCADA-Server-v1.0 (Snapshot 3) [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Oct 1 09:40
FUXA
http://localhost:1881/device
Connections settings
FUXA
Open PLC
192.168.1.10:502
Modbus-TCP Slave ID: 1



Caption: Figure 1 — FUXA Connections panel confirming active green connection status to 192.168.1.10:502 and tag table listing for Tank Level (300080) and Tank Valve Open (Offset 1).

Step 2: HMI Canvas Design / Edit Mode

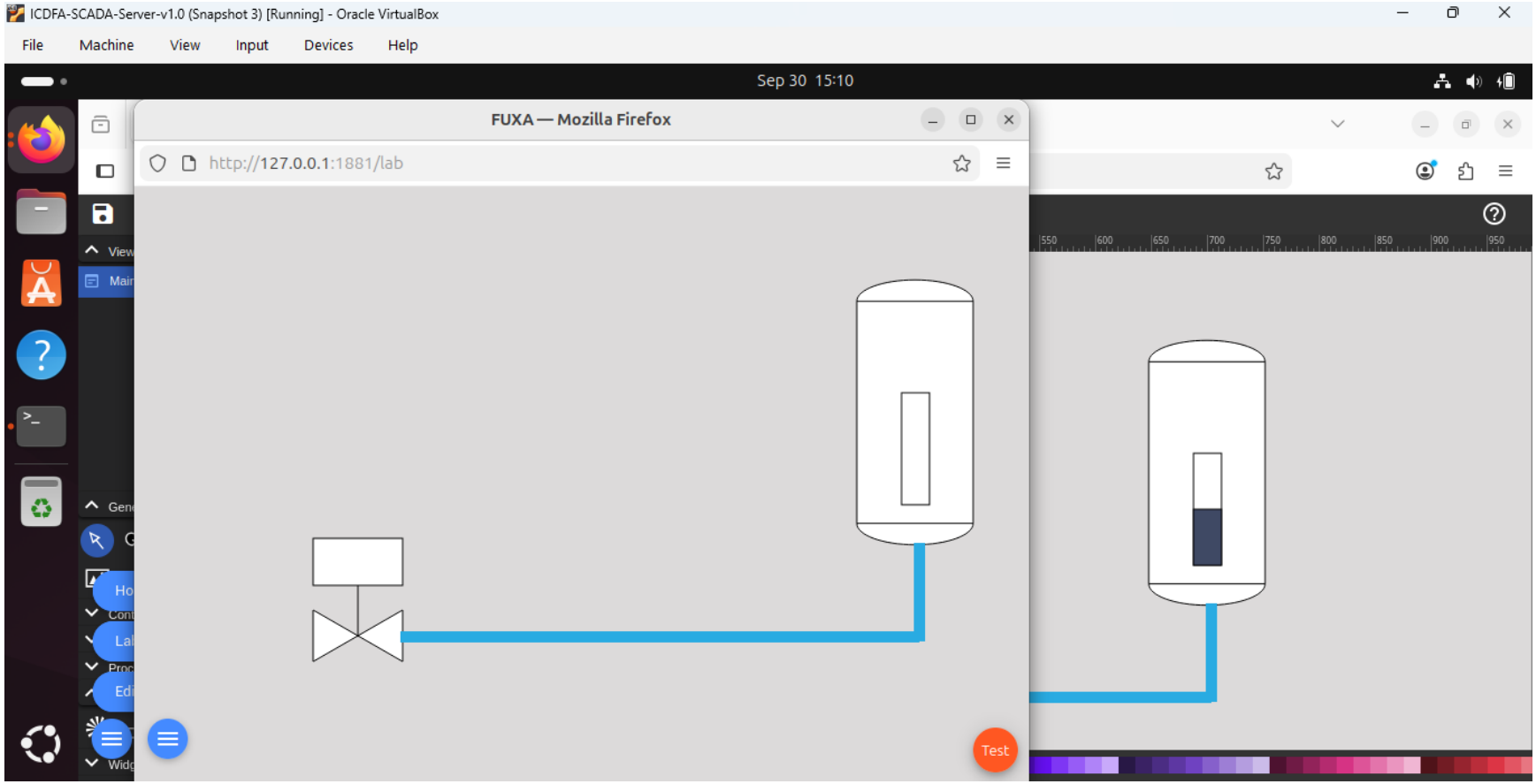
SCREENSHOT 2



Caption: Figure 2 — FUXA HMI Canvas in Editor mode (2026WT_LAB05), showing neutral gray canvas, aligned process piping, tank asset, bar gauge (0–10 ft), and valve assembly.

Step 3: Running HMI View Mode & Valve State Animation

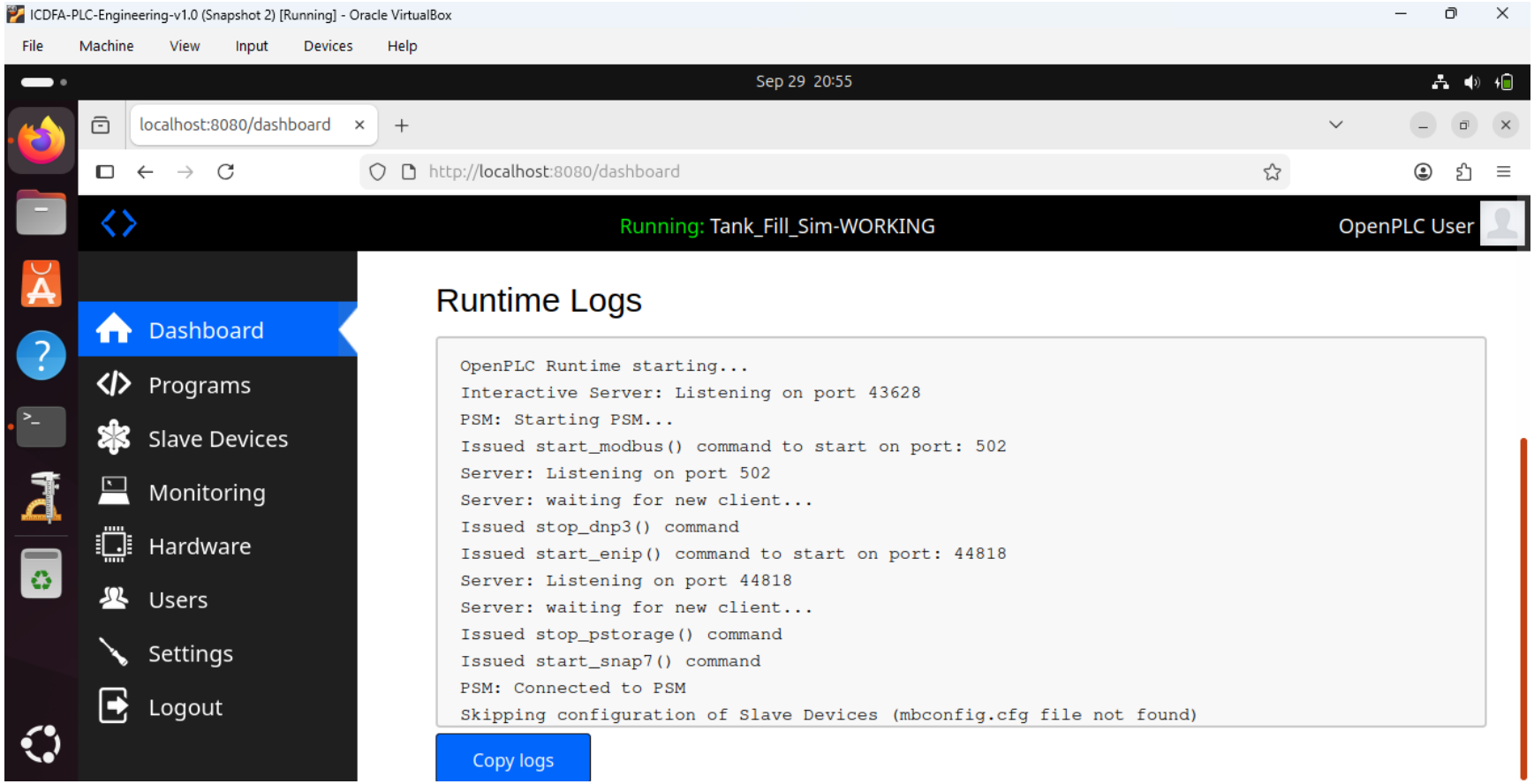
SCREENSHOT 3

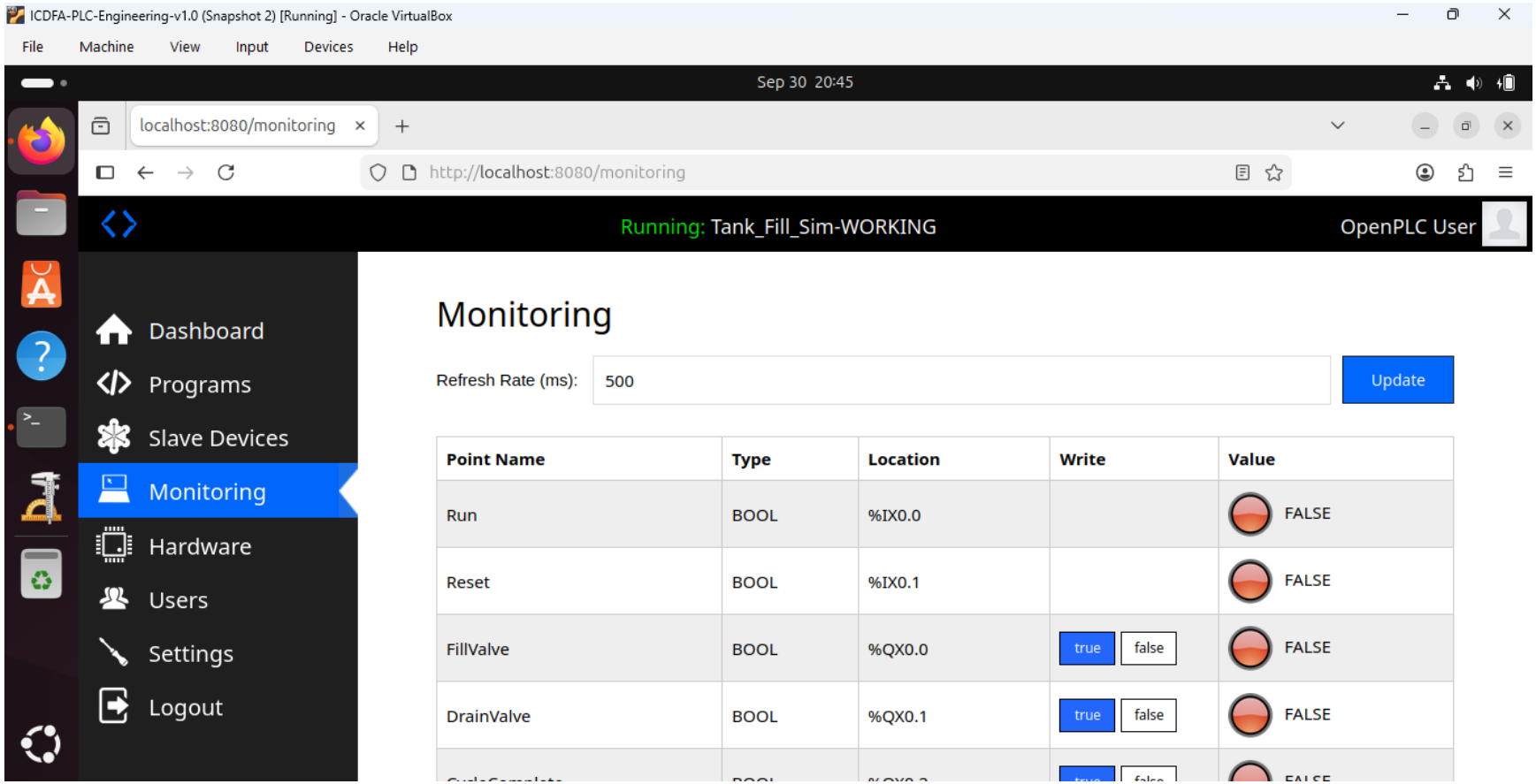


Caption: Figure 3 — Live HMI View mode demonstrating real-time response: dynamic valve graphic overlay visible when Tank Valve Open is set to 1.

Step 4: OpenPLC Runtime Monitoring & Variable Injection

[INSERT SCREENSHOT 4 HERE]


ICDFA-PLC-Engineering-v1.0 (Snapshot 2) [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Sep 29 20:55
localhost:8080/dashboard
http://localhost:8080/dashboard
Running: Tank_Fill_Sim-WORKING
OpenPLC User
Dashboard
Programs
Slave Devices
Monitoring
Hardware
Users
Settings
Logout
Runtime Logs
OpenPLC Runtime starting...
Interactive Server: Listening on port 43628
PSM: Starting PSM...
Issued start_modbus() command to start on port: 502
Server: Listening on port 502
Server: waiting for new client...
Issued stop_dnp3() command
Issued start_enip() command to start on port: 44818
Server: Listening on port 44818
Server: waiting for new client...
Issued stop_pstorage() command
Issued start_snap7() command
PSM: Connected to PSM
Skipping configuration of Slave Devices (mbconfig.cfg file not found)
Copy logs



Caption: Figure 4 — OpenPLC web monitoring screen showing FillValve (%QX0.0) write controls synchronized with FUXA test execution panel.

5. Design Rationale & Situational Awareness

The HMI layout was engineered following high-performance SCADA standards (ISA-101 / ASM Consortium):

1. **Neutral Background:** A soft, neutral gray background (#D0D0D0 / #E0E0E0) is used to minimize eye fatigue during extended operational shifts and ensure high visual contrast for process elements.
2. **Restrained Color Palette:** Color is strictly reserved for actionable process states and abnormal conditions. Standard process piping and offline equipment remain gray, while active states or alarms utilize targeted colors to draw immediate operator attention.
3. **Low Cognitive Load Layout:** Equipment assets are arranged in logical process flow order (Inlet Valve → Piping → Storage Tank). Direct embedded bar gauges provide qualitative analog level assessment at a glance without requiring operators to read raw register numbers.

6. Knowledge Check

Question 1: What is a P&ID and how does it influence HMI layout?

Response:

A Piping and Instrumentation Diagram (P&ID) is a detailed schematic diagram showing the physical relationship, process piping, control loops, instrumentation, and mechanical equipment within an industrial facility. A P&ID influences HMI layout by serving as the architectural baseline for the process flow.

However, rather than copying every minor mechanical detail from a P&ID onto the screen, a high-performance HMI simplifies the P&ID layout into a clean, functional representation. It maintains accurate topological relationships (e.g., flow direction from left to right, valve placement prior to tank inlets) while eliminating unnecessary clutter, ensuring operators can navigate the interface intuitively.

Question 2: Why do high-performance HMIs avoid unnecessary animation and excessive color?

Response:

High-performance HMIs avoid unnecessary animation (such as spinning pump blades or flashing ambient lights) and excessive bright colors to prevent visual fatigue and operator distraction. In traditional, overly colorful "rainbow" HMIs, operators become desensitized to color, making it difficult to spot critical alarm conditions quickly.

By utilizing a neutral gray color palette for normal operating conditions, bright colors (such as bright red, orange, or yellow) stand out instantly when an abnormal process state or critical alarm occurs. This design philosophy maximizes situational awareness and drastically reduces operator response times during emergency events.

Question 3: How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

Response:

The **Tank Level** tag and **Tank Valve Open** tag differ in data type, Modbus protocol functions, scaling requirements, and dynamic visual behaviors:

- **Tank Level (Analog Tag):**
  - **Data Type:** Int16 (16-bit integer register, Modbus Input Register / Holding Register).
  - **Behavior:** Represents a continuous numerical measurement (e.g., fluid height from 0 to 10 ft).
  - **HMI Mapping:** Bound to analog display elements like vertical bar gauges or numerical readouts, requiring engineering scaling (Min = 0, Max = 10).
- **Tank Valve Open (Discrete / Digital Tag):**
  - **Data Type:** Bool (1-bit boolean flag, Modbus Discrete Output / Coil).
  - **Behavior:** Represents a binary state containing only two discrete conditions: 0 (FALSE/Closed) or 1 (TRUE/Open).
  - **HMI Mapping:** Bound to visibility rules or color-toggle states (e.g., Hide graphic on 0, Show graphic on 1), requiring no numerical scaling.

Question 4: What could happen operationally if an HMI displays stale or incorrect PLC data?

Response:

If an HMI displays stale or incorrect data due to communication latency, frozen Modbus polling, or incorrect tag offset configuration, the operator loses true situational awareness. Operational risks include:

1. **Unintentional Overfills / Spills:** The HMI may display a safe fluid level (e.g., 2 ft) when the physical tank is actually overflowing, resulting in toxic chemical spills, environmental hazards, or severe facility flooding.
2. **Equipment Damage & Dry Runs:** An operator might start a high-capacity pump believing fluid is present, running the pump dry and causing catastrophic mechanical failure or overheating.
3. **Inappropriate Manual Intervention:** Operators relying on inaccurate data may manually override automated control valves or shut down critical process lines unnecessarily, leading to costly unmonitored production downtime.
4. **Safety Instrument System (SIS) Blindness:** Delayed reporting of pressure or temperature spikes prevents operators from initiating manual emergency shutdowns (ESD) before automated hardwired safety trips occur.